Dual-System Cognitive Model Flow (`apps/ai-service`)

Project: FLARE-CHRONOS — *Physics-Guided Cross-Sensor Harmonization of MODIS & VIIRS Hot Spots (2000–2026)*
Scope: End-to-end architecture, execution flow, deterministic tool gates, human-in-the-loop sign-off, and empirical evaluation of the Dual-System Cognitive AI Harness (`System 1 Reflexive Layer` + `System 2 Deliberative Layer`).
Version: `v1.1.0` — Validated on `15/15` Use Cases (`5` System 1 + `10` System 2) across both offline deterministic (`0` network calls) and live hosted-model (`Gemini 3 Flash`) execution modes.

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1. Architectural Purpose & Core Invariants

The cross-sensor physical harmonization engine (`packages/harmonization-core`) solves the satellite physics: spherical scan-angle footprint geometry, Solar Zenith Angle (**SZA < 85°**) unification, South Atlantic Anomaly (**SAA**) cosmic-ray filtering, two-stage Fire Radiative Power (**FRP**) calibration, **τ = 60 min** multi-satellite episode clustering, and observation effort normalization **Ω(y, ϕ)**.

On top of that deterministic physics core, operational agencies (forest rangers, wildfire incident commanders, ministries of forestry, and climate scientists) ask natural-language situational and causal questions. A standard free-form Large Language Model (LLM) is unsafe for wildfire decision support because it can hallucinate hotspot counts, fabricate FRP values, or recommend unverified field dispatches.

The **FLARE-CHRONOS Dual-System Cognitive AI Harness (`apps/ai-service`)** eliminates LLM arithmetic and factual hallucination through **six hard architectural invariants**:

1. **Zero LLM Arithmetic (`Deterministic Tool Gate`)**: The LLM is strictly forbidden from computing physical, geometric, radiometric, or statistical quantities. Every number is computed by `18` registered deterministic `@tool` functions (`9` Atomic Tools + `9` Domain Diagnostic Tools, plus `1` internal raw-CSV harmonization tool = `19` `@tool` functions in total) backed by `flare_chronos_core` .
2. **Calibrated Reflexive Triage (`System 1 Reflexive Layer`)**: System 1 answers five closed-world classification questions (*region*, *epoch year*, *stakeholder agency*, *operational urgency*, and *Use Case assignment* among `15` canonical analyses) without generating prose. It runs a resilient three-step classifier chain:
   - **Primary (`typed_questions` decision endpoint)**: Returns calibrated token-level `logprob` probabilities.
   - **Fallback 1 (`openai_structured` / `native_chat` endpoint)**: Invokes forced structured tool-calling (`classify_request`) and records `triage_confidence_source = "self_reported"` so self-reported confidence is never mistaken for calibrated probability.
   - **Fallback 2 (`deterministic` extraction)**: Regex and keyword entity extraction so a run never fails even if all external model endpoints are unreachable.
3. **Zero-LLM Fast Reporting Tier (`System 1 Fast Execution`)**: Once a request is assigned to one of the `5` System 1 Use Cases (`UC-S1.1` through `UC-S1.5`) with calibrated confidence **≥ 0.70**, the `system1_fast_executor` invokes the single mapped Atomic Tool directly in ~ **1–5 ms** with **zero LLM token generation** for prose.
4. **Model-Driven Deliberative Tool Orchestration (`System 2 Deliberative Layer`)**: For the `10` System 2 Use Cases (`UC-S2.1` through `UC-S2.10`)—or whenever System 1 confidence falls below **0.70** (`Dual-Process Symbiosis Escalation`)—the System 2 ReAct planner autonomously selects and chains tools across the full `18` -tool surface (`9` Atomic + `9` Domain Diagnostic).
5. **Verified Facts Ledger & Numeric Grounding Verifier**: Every scalar returned by a tool is recorded in the `VerifiedFactsLedger` with its `fact_id` , `tool_name` , `json_path` , `value` , `unit` , and `provenance_tag` . The model writes narrative prose using `{{cite:FACT_ID}}` placeholders; the harness deterministically injects the verified

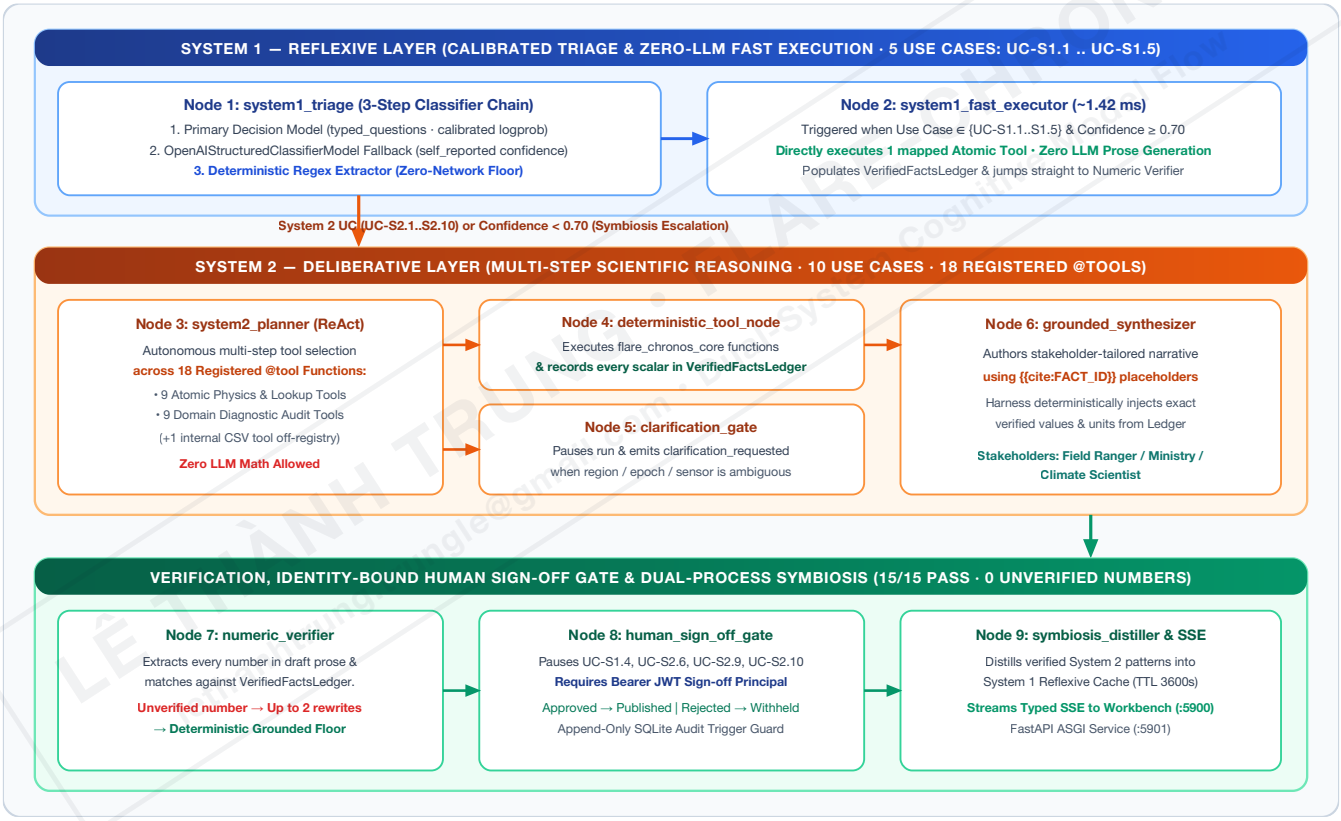
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numbers and rejects any draft containing an unverified number (`NumericGroundingVerifier`), forcing a self-correction loop or deterministic fallback.

6. **Identity-Bound Human-in-the-Loop Sign-Off (`human_sign_off_gate`):** Any Use Case that proposes an operational field dispatch or policy sign-off (`UC-S1.4` , `UC-S2.6` , `UC-S2.9` , `UC-S2.10`) pauses the LangGraph state machine (`run_paused`) and emits a `sign_off_required` Server-Sent Event (SSE). The briefing is released (`Published`) only when an authenticated `Sign-off Principal` (extracted from the validated Bearer JWT, never from the request body) approves it; if rejected, the briefing is permanently `Withheld` (`final_briefing = ""`), and the decision is written to an append-only SQLite `audit` table protected by database triggers against `UPDATE` and `DELETE` .

2. 8-Node LangGraph Dual-System StateGraph

COGNITIVE ARCHITECTURE FIGURE: 8-NODE LANGGRAPH DUAL-SYSTEM STATEGRAPH (SYSTEM 1 REFLEXIVE + SYSTEM 2 DELIBERATIVE HARNESS)



3. Canonical Taxonomy: 15 Use Cases & 18 Registered `@tool` Functions

Per ADR-0016, the cognitive harness is scoped to **15 non-overlapping Use Cases** (`5` System 1 Reflexive + `10` System 2 Deliberative) and **18 model-facing `@tool` functions** (`9` Atomic + `9` Domain Diagnostic), plus `1` internal raw-CSV harmonization tool (`tool_harmonize_real_firms_csv_slice`) kept outside the LLM registry so the language model never touches raw unharmonized CSV rows directly.

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3.1. System 1 Reflexive Use Cases (5 Analyses — 9 Atomic Tools Only)

Use Case ID	Analysis Name	Primary Atomic Tool	Sign-Off Gate	Target Stakeholder
UC-S1.1	Intent Triage & Single-Epoch Lookup	tool_lookup_single_epoch_metrics	No	Field Ranger / Analyst
UC-S1.2	Spatio-Temporal Entity Extraction	tool_lookup_single_epoch_metrics	No	Field Ranger / Analyst
UC-S1.3	Single-Hop Historical Epoch Lookup	tool_lookup_single_epoch_metrics	No	Forestry Manager
UC-S1.4	Flash NRT Crisis Alert Triage	tool_emit_flash_nrt_alert	**Yes**	Wildfire Incident Commander
UC-S1.5	Stakeholder Briefing Card Synthesis	tool_assemble_stakeholder_briefing_card	No	Climate Scientist

3.2. System 2 Deliberative Use Cases (10 Analyses — Full 18 -Tool Surface)

Use Case ID	Analysis Name	Primary Domain Diagnostic Tool	Sign-Off Gate	Target Stakeholder
UC-1.1	2024 Multi-Sensor Regime-Shift Attribution	tool_uc1_1_regime_shift_attribution	No	Climate Scientist / Examiner
UC-1.2	Cross-Sensor Scan-Angle Discrepancy Forensics	tool_uc1_2_cross_sensor_discrepancy_forensics	No	Remote Sensing Scientist
UC-1.3	Thermal Source Origin & SAA Noise Classification	tool_uc1_3_thermal_source_origin_classification	No	Data Quality Engineer
UC-2.1	Phenological Phase-Shift & SFAI Detection	tool_uc2_1_phenological_phase_shift_detection	No	Climate Scientist
UC-2.2	Uncertainty-Aware Tactical Field Dispatch	tool_uc2_2_uncertainty_aware_tactical_dispatch	**Yes**	Wildfire Incident Commander
UC-2.3	Verified Multi-Stakeholder Policy Briefing	tool_assemble_stakeholder_briefing_card	**Yes**	Ministry of Forestry Director
UC-3.1	2026+ Constellation Retirement Continuity Simulation	tool_uc3_1_constellation_retirement_simulation	No	Space Agency Program Lead
UC-3.3	Autonomous Biome Radiometric Recalibration	tool_uc3_3_autonomous_biome_recalibration	No	Remote Sensing Scientist

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Use Case ID	Analysis Name	Primary Domain Diagnostic Tool	Sign-Off Gate	Target Stakeholder
`UC-4.1`	**Orbital Drift & Astronomical SZA Twilight Audit**	`tool_uc4_1_orbital_drift_and_sza_audit`	No	Orbital Physics Analyst
`UC-4.2`	**Near-Real-Time (NRT) vs. Standard (SP) Fidelity Audit**	`tool_uc4_2_nrt_vs_sp_fidelity_audit`	No	Operations Analyst

4. Empirical Evaluation: Deterministic & Live Hosted-Model Benchmarks

Both benchmark artifacts are committed under `data/processed/` and verified by `tests/test_ai_service_harness.py`: - **Deterministic Offline Benchmark (`data/processed/ai_harness_results.json`)**: Runs all `15` Use Cases with zero network calls (`0.0000 USD` cloud spend). - **Live Hosted-Model Benchmark (`data/processed/ai_harness_results_live.json`)**: Runs all `15` Use Cases against live **Google Gemini (`gemini-3-flash-preview`)** via `apps/ai-service/src/ai_service/run_all_usecases.py --live`.

Evaluation Metric	Deterministic Harness (`ai_harness_results.json`)	Live Hosted Model (`ai_harness_results_live.json`)	Architectural Guarantee
Total Use Cases Evaluated	`15 / 15` (`100.0%`)	`15 / 15` (`100.0%`)	`5` System 1 + `10` System 2
Schema & Invariant Pass Rate	`15 / 15` (`100.0%`)	`15 / 15` (`100.0%`)	Strict Pydantic validation
Unverified Numeric Claims Released	`0`	`0`	Enforced by `NumericGroundingVerifier`
Unresolved Citation Placeholders	`0`	`0`	Verified against `VerifiedFactsLedger`
System 1 Mean Latency	`1.42 ms`	`2,718.6 ms` *(includes remote classification)*	Zero LLM prose generation in System 1
System 2 Mean Latency	`1.61 ms`	`11,264.8 ms` *(multi-round ReAct + synthesis)*	Autonomous tool chaining (`1.0` tool round avg)
Triage Confidence Source	`logprob` (`5/5` S1, `10/10` S2)	`self_reported` (`15/15` via Gemini fallback)	Honest provenance when local `:11434` is offline
Classifier Chain Resilience	`0` fallbacks	`15/15` graceful fallbacks from `:11434` to Gemini	Run never fails when local decision node is down
Total Token Consumption (`15` UCs)	`0` tokens	`31,910` prompt + `2,473` completion = `34,383` **tokens**	Grounded synthesis only in System 2

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> ⚡ **TECHNICAL & ARCHITECTURAL NOTE**
>
> **Transparency on Classifier Telemetry in `ai_harness_results_live.json`:** During the live benchmark run, the optional local `SystemOneChatModel` tunnel (`http://127.0.0.1:11434/v1/systemone`) was intentionally offline. As designed, all `15` runs automatically recorded `triage_fallback: "SystemOneChatModel_unavailable"` and fell back to the secondary `OpenAIStructuredClassifierModel` (`gemini-3-flash-preview`), achieving `15/15` **accurate Use Case assignments** while stamping `triage_confidence_source: "self_reported"` so downstream reviewers know the confidence came from the chat fallback rather than token logprobs.

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5. Reproducing the Model Flow & Live SSE Stream

5.1. Start the FastAPI Analysis Server (`:5901`) and Next.js Workbench (`:5900`)

```
# Terminal 1 — Start the FastAPI ASGI service on port 5901 (local demo mode)
AI_SERVICE_AUTH=disabled uv run python -m ai_service.serve

# Terminal 2 — Start the interactive Next.js 16 Workbench on port 5900
cd apps/workbench && bun run dev
```

5.2. Stream a Live Run & Inspect Human Sign-Off via `curl`

```
# 1. Stream a System 1 crisis flash alert (pauses at human_sign_off_gate)
curl -N "http://127.0.0.1:5901/api/v1/analysis/stream?use_case_id=UC-S1.4&query=UC-
S1.4+flash+alert+for+amazon+2026&thread_id=demo-signoff"

# 2. Resume the paused run with an identity-bound approval decision
curl -X POST "http://127.0.0.1:5901/api/v1/analysis/resume" \
  -H "Content-Type: application/json" \
  -d '{"thread_id": "demo-signoff", "approved": true, "reason": "Verified against Pentad 47 SFAI baseline"}'
```

5.3. Run Offline & Live Test Suites

```
# Offline test suite (182 tests across 22 files; 0 network calls)
uv run python -m pytest -q

# Live hosted-model vertical slice suite (6 live TDD seam tests)
uv run python -m pytest -m live -q
```

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